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Reverse flow gas turbine engine having electric machine

Abstract

An aircraft engine assembly includes a gas turbine engine having an intake channel configured to receive an incoming flow of air and thereby form an intake flow of air, the intake channel configured to turn the received incoming flow of air from an incoming flow direction to a first axial direction of the gas turbine engine, the incoming flow direction reverse of the first axial direction, and an electric machine coupled with the low pressure shaft and located at the aft end of the gas turbine engine proximate the intake channel, the electric machine in heat exchange communication with the intake flow of air such that the electric machine transfers heat to the incoming flow of air within the intake channel when the electric machine is operated.

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Background/Summary

PRIORITY INFORMATION

(1) The present application claims priority to U.S. patent application Ser. No. 18/307,938, filed on Apr. 27, 2023, which claims priority to Polish Patent Application Number P.443814, filed on Feb. 17, 2023. U.S. patent application Ser. No. 18/307,938 and Polish Patent Application Number P.443814 are hereby incorporated by reference in their entirety for all purposes.

FIELD

(2) The present disclosure relates generally to a reverse flow gas turbine engine having an electric machine.

BACKGROUND

(3) A gas turbine engine generally includes a turbomachine and a rotor assembly. Gas turbine engines, such as turboprop engines, may be used for aircraft propulsion. In the case of a turboprop engine, the rotor assembly may be configured as a variable pitch propeller. In some installations the gas turbine engine is oriented in a reverse flow configuration such that an air flow provided by forward motion of an aircraft is received by the gas turbine engine and turned to flow in a reverse direction through the turbomachinery of the gas turbine engine before an exhaust is discharged from the engine. Locating auxiliary components in such a reverse flow configuration remains an area of interest.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A full and enabling disclosure of the presently described technology, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:
- (2) FIG. 1 is a cross-sectional view of a gas turbine engine in accordance with an exemplary aspect of the present disclosure.
- (3) FIG. 2 is a schematic view of a gas turbine engine with an electric machine in accordance with the present disclosure.
- (4) FIG. 3 is a schematic view of a gas turbine engine with an electric machine in accordance with the present disclosure.
- (5) FIG. 4 is a schematic view of a gas turbine engine with an electric machine in accordance with the present disclosure.
- (6) FIG. 5 is a schematic view of a gas turbine engine with an electric machine in accordance with the present disclosure.
- (7) FIG. 6 is a view of a controller in accordance with the present disclosure.
- (8) FIG. 7A is a cross sectional view of a non-annular portion of a flow path of an intake channel in accordance with the present disclosure.
- (9) FIG. 7B is a cross sectional view of an annular portion of a flow path of an intake channel in accordance with the present disclosure.
- (10) FIG. 8 is a flow chart describing a method of operating a gas turbine engine having an electric machine in accordance with the present disclosure.
- (11) FIG. 9 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine as shown in FIG. 1, including a portion of the inlet frame and a portion of the high pressure compressor in accordance with the present disclosure.
- (12) FIG. 10 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine as shown in FIG. 1, including a portion of the inlet frame and a portion of the high pressor compressor in accordance with the present disclosure.
- (13) FIG. 11 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine as

shown in FIG. 1, including a portion of the inlet frame and a portion of the high pressure compressor in accordance with the present disclosure.

(14) FIG. 12 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine as shown in FIG. 1, including a portion of the inlet frame and a portion of the high pressure compressor in accordance with the present disclosure.

(15) FIG. 13 is an aft looking forward schematic view of a forward portion of an exemplary rotor shaft according to the present disclosure.

(16) FIG. 14 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine as shown in FIG. 1, including a portion of the inlet frame and a portion of the high pressure compressor in accordance with the present disclosure.

(17) FIG. 15 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine as shown in FIG. 1, including a portion of the inlet frame and a portion of the HP compressor in accordance with the present disclosure.

DETAILED DESCRIPTION

(18) Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

(19) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

(20) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

(21) The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.

(22) The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source.

(23) The term “combustion section” refers to any heat addition system for a turbomachine. For example, the term combustion section may refer to a section including one or more of a deflagrative combustion assembly, a rotating detonation combustion assembly, a pulse detonation combustion assembly, or other appropriate heat addition assembly. In certain example embodiments, the combustion section may include an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or other appropriate combustion system, or combinations thereof.

(24) The terms “low” and “high”, or their respective comparative degrees (e.g., -er, where applicable), when used with a compressor, a turbine, a shaft, or spool components, etc. each refer to relative pressure within an engine unless otherwise specified. For example, a “low turbine” or “low pressure turbine” defines a component configured to operate at a pressure lower than a “high pressure turbine” of the engine.

(25) The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of a vehicle such as an aircraft. For example, with regard to an aircraft, forward refers to a position closer to a nose of the aircraft and aft refers to a position closer to an empennage of the aircraft.

(26) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

(27) As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a centerline of the gas turbine engine. Moreover, the terms “radial” and

“radially” refer to directions and orientations that extend substantially perpendicular to the centerline of the gas turbine engine. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the centerline of the gas turbine engine.

(28) As will be discussed in more detail below, the subject matter of the present disclosure is directed generally to locating an electric machine near an intake channel of a reverse flow turboprop engine at a location which is closer to an aft end of the turboprop engine than to a forward end of the turboprop engine. The electric machine is rotatably coupled to a low pressure shaft of the reverse flow turboprop engine and as a consequence of the reverse flow configuration the low pressure shaft extends aft of a core of the turboprop engine. The electric machine can be operated as a generator and/or motor for use in either adding power to and/or extracting power from the low pressure shaft. Placement of the electric machine in the proximate location described above permits an exchange of heat between the electric machine and a flow of air traversing through an intake channel of the engine. Such exchange of heat can provide tighter packaging of the electric machine and/or higher heat generating operating demands placed upon the electric machine.

(29) To accommodate the placement of the electric machine in an aft location in at least one embodiment, an intake channel may be provided that forms a non-annular flow path at an inlet to the intake channel which then changes to an annular flow path around the LP shaft prior to air being delivered to a compressor of the gas turbine engine.

(30) Referring now to the drawings, wherein identical numerals indicate the same elements throughout the figures, FIG. 1 is a schematic cross-sectional view of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure. More particularly, for the embodiment of FIG. 1, the gas turbine engine is a reverse flow turboprop engine 10, referred to herein as “turboprop engine 10.” As shown in FIG. 1, turboprop engine 10 defines an axial direction A (extending parallel to a longitudinal centerline or central axis 12 provided for reference), a radial direction R, and a circumferential direction C (not shown) disposed about the axial direction A. Turboprop engine 10 generally includes a propeller section 14 and a core turbine engine 16 disposed aft of the propeller section 14 from an aircraft perspective, the propeller section 14 being operable with, and driven by, core turbine engine 16.

(31) The exemplary core turbine engine 16 depicted generally includes a substantially tubular outer casing 18 extending generally along axial direction A. Outer casing 18 generally encloses core turbine engine 16 and may be formed from a single casing or multiple casings. Core turbine engine 16 includes, in a serial flow relationship, a compressor 22, a combustion section 26, a high pressure (HP) turbine 28, a low pressure (LP) turbine 30, and an exhaust section 32. An air flow path generally extends through compressor 22, combustion section 26, HP turbine 28, LP turbine 30, and exhaust section 32 which are in fluid communication with each other.

(32) An HP shaft or spool 34 drivingly connects the HP turbine 28 to the compressor 22. An LP shaft or spool 36 drivingly connects the LP turbine 30 to propeller section 14 of the turboprop engine 10. For the embodiment depicted, propeller section 14 includes a variable pitch propeller 38 having a plurality of propeller blades 40 coupled to a disk 42 in a spaced apart manner. As depicted, the propeller blades 40 extend outwardly from disk 42 generally along the radial direction R. Each propeller blade 40 is rotatable relative to the disk 42 about a pitch axis P by virtue of the propeller blades 40 being operatively coupled to a suitable actuation member 44 configured to collectively vary the pitch of the propeller blades 40 in unison. The propeller blades 40, disk 42, and actuation member 44 are together rotatable about the longitudinal centerline 12 by LP shaft 36 across a power gear box 46. The power gear box 46 includes a plurality of gears for stepping down the rotational speed of the LP shaft 36 to a more efficient rotational fan speed and is attached to one or both of a core frame or a fan frame through one or more coupling systems. Disk 42 is covered by a rotatable front hub 48 aerodynamically contoured to promote an airflow through the plurality of

propeller blades **40**.

(33) During operation of the turboprop engine **10**, a volume of air **50** (also referred to as a free stream flow of air **51** prior to its encounter with the propeller **38**, and referred to as an incoming flow of air **50** after passage through the propeller **38**) passes through blades **40** of propeller **38** and is urged toward a radial inlet **52** of core turbine engine **16**. More specifically, turboprop engine **10** includes an intake channel **54** that defines radial inlet **52** that routes an inlet portion of air **53** of the flow of air **50** from inlet **52** downstream to compressor **22**. Though the inlet **52** is depicted as a radial inlet in the embodiment of FIG. **1**, other configurations of inlet **52** are also contemplated. For example, the inlet **52** can also take the form of an inlet arranged in an axial direction to capture the inlet portion of air **53** of the volume of air **50**. The inlet portion of air **53** of the flow of air **50** captured by the inlet **52** is referred to herein as an intake flow of air. The intake channel **54** defines the intake flow of air and generally extends from an inlet of the intake channel **54** to just upstream of the compressor **22**.

(34) The turboprop engine **10** embodiments described herein are configured as reverse flow engines. Such engines are characterized by a general relationship between the direction of the flow of incoming air **50** (such direction can be used to characterize the relative motion of air during a mode of operation of the engine **10** such as a forward thrust mode) and that of the flow of air axially through the turboprop engine **10**. The flow of air through the core turbine engine **16** is generally reverse to that of the flow of incoming air **50**. Turning the flow from the direction of the incoming flow of air **50** to the axial direction through the core turbine engine **16** is usually performed by the intake channel **54**. The change of direction is reversed in that the bulk direction of the flow of air **50** (itself having a circumferential swirl component imparted by the propeller blades **40** in addition to a longitudinal component) is opposite, or reverse, to the bulk direction of air flow axially through the core turbine engine **16** (which itself also includes a longitudinal component but also include radial and circumferential components owing to the shape of the flow path and swirl induced by rotating turbomachinery components) during one or more phases of operation of the core turbine engine **16**. Thus, it will also be appreciated that the term “reverse” is a relative comparison of the longitudinal components of the bulk flow of air **50** and bulk flow of air axially within the engine **10**. Though the longitudinal direction of the flow of air **50** may not be perfectly parallel with the axial flow of air through the engine **10**, it will be appreciated that the longitudinal components of the directions the flow of air **50** and the axial flow are reversed.

(35) Compressor **22** includes one or more sequential stages of compressor stator vanes **60**, one or more sequential stages of compressor rotor blades **62**, and an impeller **64**. Though the illustrated embodiment includes both axial and centrifugal flow compressors, in some forms the turboprop engine **10** can include just an axial flow compressor(s) or centrifugal flow compressor(s). The one or more sequential stages of compressor stator vanes **60** are coupled to the outer casing **18** and compressor rotor blades **62** are coupled to HP shaft **34** to progressively compress the air **53**. Impeller **64** further compresses air **53** and directs the compressed air **53** into combustion section **26** where air **53** mixes with fuel. Combustion section **26** includes a combustor **66** which combusts the air/fuel mixture to provide combustion gases **68**.

(36) Combustion gases **68** flow through HP turbine **28** which includes one or more sequential stages of turbine stator vanes **70** and one or more sequential stages of turbine blades **72**. The one or more sequential stages of turbine stator vanes **70** are coupled to the outer casing **18** and turbine blades **72** are coupled to HP shaft **34** extract thermal and/or kinetic energy therefrom. Combustion gases **68** subsequently flow through LP turbine **30**, where an additional amount of energy is extracted through additional stages of turbine stator vanes **70** and turbine blades **72** coupled to LP shaft **36**. The energy extraction from HP turbine **28** supports operation of compressor **22** through HP shaft **34** and the energy extraction from LP turbine **30** supports operation of propeller section **14** through LP shaft **36**. Combustion gases **68** exit turboprop engine **10** through exhaust section **32**.

(37) It will be understood that one or more rows of stator vanes **60** and **70** can be variable vanes

controlled by a controller (see below with respect to controller **100**) in one form. Furthermore, with particular respect to stator vanes **70**, one or more rows of the stator vanes **70** can be variable.

(38) In other exemplary embodiments, the turbine engine may include any suitable number of compressors, turbines, shafts, etc. For example, as will be appreciated, HP shaft **34** and LP shaft **36** may further be coupled to any suitable device for any suitable purpose. For example, in certain exemplary embodiments, turboprop engine **10** of FIG. **1** may be utilized in aeroderivative applications. Additionally, in other exemplary embodiments, turboprop engine **10** may include any other suitable type of combustor, and may not include the exemplary reverse flow combustor depicted.

(39) The embodiment of turboprop engine **10** illustrated in FIG. **1** includes an electric machine **74** located aft of the core turbine engine **16** and rotatably coupled to the LP shaft **36**. In some forms the electric machine **74** is contained in an environmentally sealed housing which can be pressurized to minimize electrical corona and discharge effects. Further, in some forms the LP shaft **36** and electric machine **74** can be configured to rotate at a constant speed from idle to max power, with thrust of the engine **10** controlled by the variable pitch propeller **38**.

(40) Given the coaxial relationship of the LP shaft **36** with the HP shaft **34**, in such an embodiment the LP shaft **36** is configured to extend aft of the core turbine engine **16**, and also further aft than the HP shaft **34**, despite the HP compressor **22** being the upstream-most compressor of the turboprop engine **10** (i.e., despite there being no low-pressure compressor upstream of the HP compressor **22** and downstream of the inlet **52**).

(41) The electric machine **74** can be used in many different power configurations. In one form the electric machine is configured to extract power from the LP shaft **36** when the machine **74** operates as a generator. The extraction of mechanical power from the LP shaft **36** and conversion to electric power can be used to charge an on-board power storage device such as a battery, or alternatively to provide power to another electrical device (e.g., an electric motor, an electrical accessory on an aircraft, etc.). In other forms, the electric machine **74** can be used as a motor to provide power to the LP shaft **36** to supplement power extracted by the LP turbine **30** from the combustion gases **68**. In these forms, the electric machine **74** can be configured to provide a minimum of 10% of supplemental thrust to the engine **10**, a minimum of 20% of supplemental thrust to the engine **10**, and up to 40% of supplemental thrust to the engine **10** in various embodiments. In still other forms, the electric machine **74** can be configured to power to drive 100% of thrust from the propeller section **14**. A scenario in which the electric machine **74** provides all power to the propeller section **14** can include shutdown of the engine **10**. In one non-limiting example of an engine being shut down, upon or near landing the engine **10** can be command to shut down and the electric machine **74** used to drive further propeller thrust requirements, whether that includes fine power on short-final or power when the propeller section **14** is configured in reverse pitch to aid in slowing the aircraft.

(42) A battery or other secondary power source can be used to provide power to the electric machine **74** when operated as a motor. The supplementation of power by the electric machine **74** to the LP shaft **36** in this manner can be transitory or steady state, depending on the control requirements requested of the turboprop engine. For example, in those operating conditions in which power output of the engine lags behind a commanded power, the electric machine **74** can provide near instantaneous supplemental power to the LP shaft **36** to provide on-condition power output from the turboprop until the engine **10** achieves a steady state operating condition at the higher output power. In still further forms, the electric machine can be operated as a motor in some portions of operation of the engine **10**, and as a generator in other portions of operation of the engine **10**, along the lines of any of the variations discussed herein.

(43) Given the proximity of the electric machine **74** to the intake channel **54**, the electric machine **74** can be further positioned to exchange heat with the inlet portion of the air **53** traversing the intake channel **54** to aid in removing heat from the electric machine **74**. Cooling of the electric

machine **74** using the inlet portion of air **53** permits tighter packaging of the electric machine **74** and closer location of the electric machine **74** to heat generating portions of the engine **10**. The relative location of the electric machine **74** and intake channel **54** can permit an effective exchange of heat through any number of useful mechanisms including at least one of conduction and convection cooling. In some forms cooling air may directly impinge upon the electric machine **74** and/or may be used to vent a cavity in which the electric machine **74** is located. In still further forms the electric machine **74** can be used in part to form the flow path of the intake channel **54** to provide direct heat transfer between the electric machine **74** and the inlet portion of air **53**. In yet still further forms the electric machine **74** may be in direct contact with a portion of the intake channel **54** forming the flow path such that heat transfer occurs between the electric machine **74** and inlet portion of air **53** via that particular portion of the intake channel **54** forming the flow path. Further aspects of the location of electric machine **74** and various cooling techniques are described further below.

(44) Various other embodiments are disclosed further herein related to the location and use of the electric machine **74**, the shape and configuration of the intake channel **54**, and various techniques to exchange heat between the electric machine **74** and the inlet portion of air **53** flowing in the intake channel **54**. As will be appreciated, like reference numerals refer to like elements and, thus, any of the variations disclosed herein related to any particular exemplary embodiment in any given figure are also applicable to embodiments depicted and discussed with respect to the other figures.

(45) Turning now to FIG. **2**, an exemplary embodiment of the turboprop engine **10** is illustrated in which the engine **10** includes an electric machine **74** located aft of the core turbine engine **16** in a tail cone **76** defining an aft end of an engine nacelle **78**. As will be appreciated, the engine nacelle **78** is used to enclose the gas turbine engine and includes one or more portions that interface with aircraft structure such as a wing, pylon, fuselage, etc. The tail cone **76** may be a complete body of revolution that circumferentially encloses the electric machine **74** in some embodiments. In alternative embodiments, however, the tail cone **76** may be a partial body of revolution or other shape that covers the electric machine to complete an enclosure with other aircraft structure (e.g., wing, pylon, fuselage, etc.). Thus, the tail cone **76** is any suitable structure of the engine nacelle which is located aft of the core turbine engine **16** and is used to wholly or partially enclose the electric machine **74**.

(46) In some forms of the embodiments disclosed herein the engine nacelle **78** may take on the form of an engine cowling when the engine **10** is installed on a single engine turboprop aircraft. In such installations the electric machine **74** can be located between the core turbine engine **16** and a firewall of the aircraft. In such installations, therefore, the engine cowling or other forebody structure of the aircraft can be used to enclose the electric machine **74**.

(47) The embodiment of FIG. **2** includes an accessory gear box (AGB) **80** located aft of the HP compressor **22** and is coupled to a starter motor **82**. The starter motor **82** is coupled to the HP shaft via the AGB **80** such that during a start sequence of the turboprop engine **10** the starter motor **82** can be used to impart rotational power via the AGB **80** to the HP shaft **34**. In the illustrated embodiment, the AGB **80** is depicted as being co-axial with the HP shaft **34** and LP shaft **36** (it will be appreciated that although the AGB **80** is rotatably coupled to the HP shaft **34**, it is not otherwise rotatably coupled with the LP shaft **36**). To provide such a coaxial relationship between the HP shaft **34** and AGB **80**, in one form the AGB **80** is a planetary gear system in which the HP shaft **34** is coupled to a sun gear of the planetary gear system. In other forms a central gear of the AGB **80** is coupled via one or more idler gears to the starter motor **82**. Other forms are also contemplated to permit a co-axial relationship between the AGB **80** and HP shaft **34**. In other forms, however, the AGB **80** need not be co-axial with the HP shaft **34**.

(48) The electric machine **74** is also depicted in FIG. **2** as being coupled to the LP shaft **36** through a speed change device **84** which can be used to alter a speed ratio between the LP shaft **36** and the electric machine **74**. In some forms, the speed change device **84** can be a transmission that provides

a fixed speed ratio, but in other forms the transmission can provide variable speed ratios. In still other forms, the transmission can include a clutch mechanism to disengage the electric machine **74** from the LP shaft **36**. Still further, although the embodiment depicted in FIG. **2** includes the speed change device **84**, other embodiments need not include the device **84**. In such an embodiment, the electric machine **74** is directly connected to the LP shaft **36**.

(49) The intake channel **54** forms a flow path between the AGB **80** and the HP compressor **22**. The flow path of the intake channel **54** follows the route depicted in FIG. **2**, where the inlet portion of air **53** captured by the intake channel **54** passes first through a non-annular inlet **90** depicted at sightline A-A (an embodiment of which is illustrated in FIG. **7A**, discussed in more detail below) before it is split into an annular flow path **92** depicted at sightline B-B (an embodiment of which is illustrated in FIG. **7B**, discussed in more detail below). The non-annular inlet **90** is radially offset from the LP shaft **36** and confined to a circumferential section about the LP shaft **36**. The intake channel **54** takes the form of the annular shape leading up to the delivery of the inlet portion of air **53** to the HP compressor **22**. The intake channel **54** thus begins with a non-annular intake shape and ends with an annular shape. As will be appreciated, the annular shape of the intake channel includes a central interior that accommodates the LP shaft **36**. The intake channel **54** in the illustrated embodiment is thus required to morph, or change, from the non-annular shape at the inlet **90** to the annular shape at its discharge to the compressor **22** so that the intake channel **54** accommodates the intrusion of the LP shaft **36** through the intake channel **54** and to the electric machine **74**. In this way, the LP shaft **36** extends through at least a portion of the intake channel **54** where the flow path in the intake channel **54** changes shape to accommodate the LP shaft **36**. In some forms, an exterior surface of the LP shaft **36** may be exposed to the inlet portion of air **53** such that the inlet portion of air **53** flowing in the intake channel **54** is exposed to the rotating exterior surface of the LP shaft **36**. In other forms, however, the intake channel **54** may include a separate structure used to define the central interior and which is used to separate the exterior surface of the LP shaft **36** from the inlet portion of air **53** flowing in the intake channel **54**.

(50) In some forms, the inlet **90** of the intake channel **54** is located at either a six o'clock position on the nacelle such as what would conventionally be considered the bottom, or underside, of the nacelle. An inlet on a Beechcraft Denali or Beechcraft King Air Turboprop are examples. The exhaust section **32** can be located at either or both of the three o'clock and nine o'clock position on the nacelle such as would conventionally be considered a left or right side of the nacelle. In this manner, the spacing of the inlet **90** of the intake channel **54** is circumferentially displaced from the exhaust section **32** to minimize/prevent exhaust gases from being circulated to the inlet **90** for ingestion into the engine **10**. Furthermore, it will be appreciated that the inlet **52**, though illustrated at an axially aft location in the various embodiments, can be located forward closer to the blades **40** while still maintaining the configuration to reverse the flow from the direction of the incoming flow of air **50** to the axial flow direction required in the turbine engine configurations depicted.

(51) Also depicted in FIG. **2** is an offtake flow path **86** created by an offtake opening **88** provided in the intake channel **54** and which is configured to provide a flow of offtake air **94** to be used for heat exchange purposes with the electric machine **74**. The offtake opening **88** can be located downstream of the inlet **90** to the intake channel **54** and is structured to remove part of the inlet portion of air **53** flowing through the offtake channel **86**. The offtake opening **88** can be a permanent vent structure that includes a fixed opening through which air can pass regardless of mode of operation of the electric machine **74**. In other forms, however, the offtake opening **88** can include a movable mechanical structure that permits modulating the area of the offtake opening **88**, including in some forms fully closing the offtake opening. Such a movable mechanical structure can take any variety of forms such as a hinged plate, sleeve valve, or other suitable device.

(52) The offtake flow path **86** can take a variety of forms including the solid line depicted in FIG. **2** in which the flow path **86** flows past the electric machine **74**. In one form, the offtake flow path **86** can alternatively and/or additionally be routed along the dotted line shown in FIG. **2**. Such a flow

path can provide additional level of venting of the cavity in which the electric machine **74** is located, and/or provide greater dwell time within the cavity to ensure a higher level of heat exchange.

(53) The offtake flow path can include one or more discharge openings **96** and **98** to permit the flow of offtake air **94** to exit from the nacelle **78**. The flow of offtake air **94** can be urged to exit through a pressure differential that exists between the offtake opening **88** and the discharge openings **96** and/or **98**. Such a pressure differential can be provide via ejector action if needed through suitable structure configured to provide such an action (e.g., a venturi ejector). One or both of the discharge openings **96** and **98**, in some embodiments, can be a permanent vent structure that includes a fixed opening through which air can pass regardless of mode of operation of the electric machine **74**.

(54) In other forms, however, one or both of the discharge openings **96** and **98** can include a movable mechanical structure that permits modulating the area of the discharge openings **96** and **98**, including in some forms fully closing the discharge openings. Such a movable mechanical structure can take any variety of forms such as a hinged plate, sleeve valve, or other suitable device. The moveable mechanical structure can protrude into a passing flow of air to which the discharged flow of offtake air **94** is being discharged, and in other forms can protrude into the offtake flow path **86**.

(55) Given the proximity of the electric machine **74** to the intake channel **54** and the configurations disclosed herein, various cooling techniques are contemplated with respect to the various embodiments. For example, though the passing flow of offtake air **94** is illustrated in FIG. 2 passing adjacent and/or around the electric machine **74**, in some embodiments, the passing flow of offtake air **94** can be directed to impinge directly upon a portion of the electric machine. Further, a surface of the electric machine **74**, such as an outer housing, can form part of the flow path of the offtake flow path **86**. In other forms, a structure forming the offtake flow path **86** can be in heat conductive relationship with a portion of the electric machine **74** (e.g., a housing of the electric machine **74**).

(56) The embodiment depicted in FIG. 2 also includes a controller **100** configured to control various aspects of the depicted embodiment (the embodiment depicted in FIG. 1 can also include a controller for control of analogous features). As depicted through the various dotted lines, the controller **100** can control one or more different systems associated with operation of the engine **10**. The dotted nature depicted in the figure denotes the optional inclusion of one or more, or all, of the systems connected with the controller **100**. For example, the controller **100** can be used to control variable stator vanes **70** in either or both turbines **28** and **30**. Additionally and/or alternatively, the controller **100** can be used to control variable stator vanes **60** in the compressor **22**. Additionally and/or alternatively, the controller **100** can be used to control fuel flow to the combustion section **26**. Additionally and/or alternatively, the controller **100** can be used to control one or more of the openings of the openings **88**, **96**, and **98**. Additionally and/or alternatively, the controller **100** can be used to control operation of the electric machine **74**. Additionally and/or alternatively, the controller **100** can command fuel flow to the combustion section **26** to be stopped and also simultaneously command the propeller section **14** to be positioned in a forward or reverse pitch configuration.

(57) The controller **100** can thus be used in any or all of the following examples. The controller **100** may selectively drive the electric machine **74** as a generator. In those situations in which the electric machine **74** is 'powered on' to operate as a generator, the controller **100** can make adjustments to fuel rate delivery to the combustion section **26** while also optionally changing position of variable stator vanes **60** and/or **70**. The controller **100** can optionally operate a clutch in the speed change device **84**.

(58) Turning now to FIG. 3, another embodiment of the reverse flow engine **10** discussed above is illustrated. The exemplary engine **10** of FIG. 3 may be configured in substantially the same manner

as the exemplary engine **10** of FIG. 2, and as such the same or similar numbers may refer to the same or similar parts.

(59) For the embodiment of FIG. 3, the engine **10** includes an electric machine **74** coupled to the engine **10** and operated as a generator, the electric machine **74** further in electrical communication via a power conduit **102** with an electric machine **74b** which is operated as a motor. The electric machine **74b** operated as a motor is used to drive a set of propeller blades **40b** apart from the blades **40a** driven by the turbine engine **10** depicted at the top of FIG. 3. The electric machine **74b** configured as a motor can be used to provide additional thrust output beyond that provided by the propeller blades **40a** driven by the turbine engine **10**.

(60) Although the electric machine **74b** is illustrated apart from any turbomachinery components such as those at the top of the figure, in some embodiments the electric machine **74b** can be integrated with a gas turbine engine much in the same manner as the turboprop engine **10** depicted at the top of the figure. In these embodiments, the electrical coupling between the electric machines **74a** and **74b** can be used to exchange power between the two (e.g., where one machine is a motor and the other a generator) or can be coupled to a common energy storage device (e.g. a bank of batteries). Any of the variations in the embodiments discussed above are also applicable to the embodiments shown in FIG. 3, such as but not limited to heat exchange between the electric machine **74** and the inlet portion of air **53**, the offtake flow path **86**, etc.

(61) Turning now to FIG. 4, another embodiment of the reverse flow engine **10** discussed above is illustrated. The exemplary engine **10** of FIG. 4 may be configured in substantially the same manner as the exemplary engine **10** of FIG. 2, and, as such, the same or similar numbers may refer to the same or similar parts.

(62) For the embodiment of FIG. 4, the engine **10** includes an electric machine **74** coupled to the HP shaft **34** of the engine **10**, in which the configuration can employ the electric machine **74** as either a motor (e.g., to start the engine **10**) or a generator (e.g., to scavenge power for electric power generation). As will be appreciated in this embodiment, the LP shaft **36** need not be extended to the rear of the engine **10**.

(63) Turning now to FIG. 5, another embodiment of the reverse flow engine **10** discussed above is illustrated. The exemplary engine **10** of FIG. 5 may be configured in substantially the same manner as the exemplary engine **10** of FIG. 2, and as such the same or similar numbers may refer to the same or similar parts.

(64) For the embodiment of FIG. 5, the engine **10** includes electric machines **74a** and **74b**, each coupled with respective LP shaft **36** and HP shaft **34**. This embodiment enables power input/extraction to/from either spool **34,36** independently, as well as the potential to transfer power between the two spools **34,36** of the engine to improve operability/mitigate vibration, compressor stall or instability issues. As will therefore be appreciated, power can be extracted from spool **34** and provided to spool **36** in one mode of operation, power can be extracted from spool **36** and provided to spool **34** in another mode of operation, and power can be either extracted from or provided to both spools **34,36** in yet another mode of operation.

(65) Any of the electric machines discussed herein are capable of being packaged so as to provide power densities suitable for use in the applications discussed herein. For example, the electric machines can have a power density ranging anywhere from greater than 3 kW/kg, greater than 5 kW/kg, and greater than 6 kW/kg.

(66) As noted, the exemplary controller **100** useful in any of FIGS. 2-5 is configured to regulate any of the aforementioned systems such as variable stator vanes **60** and/or **70**, electric machine **74**, fuel flow to the combustion section **26**, etc., either alone or in combination, based on a control scheme stored in the controller **100**. In one or more exemplary embodiments, the controller **100** depicted in FIGS. 2 and 3 may be a stand-alone controller **100** for any of the aforementioned systems, or alternatively, may be integrated into one or more of a controller for the gas turbine engine with which the aforementioned systems are integrated, a controller for an aircraft including

the gas turbine engine with which the aforementioned systems are integrated, etc.

(67) Referring particularly to the operation of the controller **100**, in at least certain embodiments, the controller **100** can include one or more computing device(s) **104** such as depicted in FIG. **6**. The computing device(s) **104** can include one or more processor(s) **104A** and one or more memory device(s) **104B**. The one or more processor(s) **104A** can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, and/or other suitable processing device. The one or more memory device(s) **104B** can include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, and/or other memory devices.

(68) The one or more memory device(s) **104B** can store information accessible by the one or more processor(s) **104A**, including computer-readable instructions **104C** that can be executed by the one or more processor(s) **104A**. The instructions **104C** can be any set of instructions that when executed by the one or more processor(s) **104A**, cause the one or more processor(s) **104A** to perform operations. In some embodiments, the instructions **104C** can be executed by the one or more processor(s) **104A** to cause the one or more processor(s) **104A** to perform operations, such as any of the operations and functions for which the controller **100** and/or the computing device(s) **104** are configured, the operations for any of the aforementioned systems such as variable stator vanes **60** and/or **70**, electric machine **74**, fuel flow to the combustion section **26**, etc., as described herein, and/or any other operations or functions of the one or more computing device(s) **104**. The instructions **104C** can be software written in any suitable programming language or can be implemented in hardware. Additionally, and/or alternatively, the instructions **104C** can be executed in logically and/or virtually separate threads on the one or more processor(s) **104A**. The one or more memory device(s) **104B** can further store data **104D** that can be accessed by the one or more processor(s) **104A**. For example, the data **104D** can include data indicative of power flows, data indicative of engine/aircraft operating conditions, and/or any other data and/or information described herein.

(69) The computing device(s) **104** can also include a network interface **104E** used to communicate, for example, with the other components of system (e.g., via a communication network). The network interface **104E** can include any suitable components for interfacing with one or more network(s), including, for example, transmitters, receivers, ports, controllers, antennas, and/or other suitable components. One or more devices can be configured to receive one or more commands from the computing device(s) **104** or provide one or more commands to the computing device(s) **104**.

(70) The network interface **104E** can include any suitable components for interfacing with one or more network(s), including, for example, transmitters, receivers, ports, controllers, antennas, and/or other suitable components.

(71) The technology discussed herein makes reference to computer-based systems and actions taken by and information sent to and from computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel.

(72) Turning now to FIGS. **7A** and **7B**, cross sectional views of two separate flow stations along the flow path of the intake channel **54** are illustrated. FIG. **7A** illustrates view A-A from FIGS. **2** and **3**, above, which depicts a cross section of the intake channel **54** that routes an inlet portion of air **53** of the flow of air **50**. The cross-section A-A depicted in FIG. **7A** is at or close to the inlet of the intake channel **54** and has a non-annular flow shape. The non-annular flow shape depicted in FIG. **7A** is similar to a kidney shape in the illustrated embodiment, but other non-annular shapes are also

contemplated. The cross-section B-B depicted in FIG. 7B, from FIGS. 2 and 3 above, is at or close to the discharge of the intake channel **54** as it delivers the inlet portion of air **53** of the flow of air **50** to the compressor **22**. The annular flow shape depicted in FIG. 7B encloses the LP shaft **36**.

(73) Referring now to FIG. 8, a flow diagram of a method of operating a gas turbine engine in accordance with an exemplary aspect of the present disclosure is provided. The method of FIG. 7 may be utilized to operate one or more of the exemplary engine and electric machine placement described above with reference to FIGS. 1 through 5B. Accordingly, it will be appreciated that the method may generally be utilized to operate a gas turbine engine including engine **10**, electric machine **74**, and intake channel **54**. However, in other exemplary aspects, the method may additionally or alternatively be utilized to operate any other suitable gas turbine engine.

(74) More specifically, FIG. 8 discloses a method **106** of operating a reverse flow gas turbine engine which includes at **108** operating a gas turbine engine having a reverse flow configuration. The engine can take the form of a turboprop as discussed above in various embodiments. Step **110** includes receiving an incoming flow of air into an intake channel of the gas turbine engine. The flow of air received in the intake channel is turned at step **112** from its initial flow direction into an axial flow direction of the gas turbine engine. Step **114** discloses cooling an electric machine located aft of a turbine of the gas turbine engine using the flow of air in the intake channel.

(75) The arrangement of the electric machine **74** coupled with the low pressure shaft **36** and located on a side of the high pressure compressor **22** opposite the high pressure turbine **28** provides various technical effects, including the ability to cool the electric machine **74** using the intake flow of air. Such a placement provides for additional separation from hot section components of the engine **10**. The electric machine **74** can be placed in proximity to the intake channel **54** at an aft end of the engine **10** in a heat exchange relationship, where the intake flow of air can exchange heat through either or both of conduction and convection with the electric machine **74**. In some forms an offtake flow of air can be extracted from the intake flow of air for use in cooling the electric machine **74** in lieu of the intake flow of air for additional flexibility, such as selective cooling provided through activation of discharge openings **96,98**. Various other flexible arrangements can also be provided of the electric machine **74**, and specifically cooling of the electric machine. For example, an electric machine **74b** can be coupled with the low pressure shaft **36**, while another electric machine **74a** is coupled with the high pressure shaft **34**. Both of electric machines **74a** and **74b** can be located on a side of the high pressure compressor **22** opposite the high pressure turbine **28**.

(76) It will further be appreciated that, in certain embodiments, as will be discussed in more detail below, certain embodiments provide an inlet guide vane to a compressor of the gas turbine engine (e.g., the gas turbine engine **10**). Inlet guide vanes are statically mounted to a static structural portion of the engine and are non-rotating or circumferentially stationary with respect to a rotating shaft of the gas turbine engine during engine operation. Inlet guide vanes condition or guide airflow into the compressor.

(77) During operation, the compressor inlet guide vanes (IGVs) may be exposed to below freezing environments which may result in ice formation and buildup on or around the inlet guide vanes, particularly along an inner radial surface of the inlet where the engine inlet is curved, and the inlet guide vane is coupled to an inner surface of the inlet. Ice buildup at the inlet may be non-uniform due to engine operating conditions and/or the shape of the engine inlet. Additionally, when a heated part is present upstream of the vanes, ice crystal icing risk is elevated.

(78) The present disclosure provides an inlet guide vane that is cantilever mounted to an outer casing and extends radially inwardly towards an outer surface of a forward end portion of a rotor shaft. A tip portion of the inlet guide vane extends radially inwardly towards but does not touch the outer surface of the rotor shaft so as to form a radial gap therebetween. Ice formation is inhibited or mitigated by allowing the rotor shaft to spin beneath the tip portion of the inlet guide vane.

(79) In certain embodiments, purge air is introduced directly into the radial gap and impinges on the tip portion of the inlet guide vane thereby reducing or preventing the formation of ice along the

inlet guide vane particularly near the tip portion upstream from an inlet to the compressor. In other embodiments, purge air is introduced upstream from the radial gap flows against the leading edge of the inlet guide vane proximate to the radial gap, reducing or preventing the formation of ice along the inlet guide vane and/or an inner surface of an inlet frame upstream from an inlet to the compressor. In other embodiments, the inlet guide vane includes internal air passages and one or more outlets defined along the tip portion and/or along the leading edge portion of the inlet guide vane.

(80) FIG. 9 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine 10 as shown in FIG. 1, including a portion of the intake channel 54, also referred to as the inlet frame 54, and a portion of the HP compressor 22 in accordance with an embodiment of the present disclosure. As shown in FIG. 9, a forward end portion 170 of the HP shaft 34 defines an outer surface 172. A first stage or first row 174 of compressor rotor blades 154 (only one shown in FIG. 9) is coupled to the HP shaft 34 downstream from the forward end portion 170. An outer casing 176 such as a portion of the outer casing 18 and/or a portion of the inlet frame 54 at least partially surrounds the first row 174 of compressor rotor blades 154 and the outer surface 172 of the forward end portion 170 of the HP shaft 34. The outer casing 176 at least partially defines a compressor inlet 178 to the HP compressor 22.

(81) In certain embodiments, an inlet guide vane 100 is positioned upstream from the first row 174 of compressor rotor blades 154. The inlet guide vane 100 is not rotatable about the longitudinal centerline or central axis 12 of the turboprop engine 10. It is to be understood that although only one inlet guide vane 100 is shown in FIG. 9, the turboprop engine 10 may include a plurality of inlet guide vanes 100 annularly arranged about the longitudinal centerline or central axis 12 and the forward end portion 170 of the HP shaft 34. As shown in FIG. 9, the inlet guide vane 100 includes or defines a base or mounting portion 102, a vane body 104 extending radially inwardly with respect to radial direction R from the mounting portion 102, a tip portion 106 radially spaced from the mounting portion 102, a leading-edge portion 108, and a trailing-edge portion 110 defined downstream from the leading-edge portion 108.

(82) The mounting portion 102 is coupled to the outer casing 176 upstream from the first row 174 of compressor rotor blades 154. The inlet guide vane 100 may be rotatable about a respective blade axis “BA” or may be fixed in position. In the embodiment shown in FIG. 9, the tip portion 106 extends towards but does not touch the outer surface 172 of the forward end portion 170 of the HP shaft 34 so as to define a radial gap 112 between the tip portion 106 and the outer surface 172 of the forward end portion 170 of the HP shaft 34. In this mounting configuration, the inlet guide vane 100 is only fixed or constrained at the mounting portion 102 or in other words, it is cantilevered from the outer casing 176. In the embodiment shown in FIG. 9, the forward end portion 170 of the HP shaft 34, particularly the outer surface 172, is allowed to spin beneath the tip portion 106 of the inlet guide vane 100, thereby breaking up and/or preventing ice buildup within the radial inlet 52 at the inlet guide vane 100 upstream from the compressor inlet 178.

(83) In one exemplary embodiment as shown in FIG. 9, an axial gap or aperture 114 is defined between the forward end portion 170 of the HP shaft 34 and an inner surface or inner wall 180 of the inlet frame 5 upstream from the inlet guide vane 100. In this embodiment, the forward end portion 170 of the HP shaft 34 terminates axially forward of the leading-edge portion 108 of the inlet guide vane 100. The aperture 114 is in fluid communication with a purge or extraction-air source 182 such as the HP compressor 22. The aperture 114 is oriented to direct a flow of purge—or heated compressed air, indicated by arrows 184, towards the radial gap 112. In this embodiment, the forward end portion 170 of the HP shaft 34, particularly the outer surface 172, is allowed to spin beneath the tip portion 106 of the inlet guide vane 100 and the heated compressed air 184 is directed towards and drawn into the radial gap 112, breaking up and/or preventing ice buildup within the radial inlet 52 at the inlet guide vane 100 upstream from the compressor inlet 178.

(84) FIG. 10 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine 10

as shown in FIG. 1, including a portion of the inlet frame 54 and a portion of the HP compressor 22 in accordance with certain embodiments of the present disclosure. The various components shown in FIG. 10 are similar to or the same as those shown in FIG. 9 and described herein and are numbered likewise. As shown in FIG. 10, a seal body 186 can be disposed and extend radially between the tip portion 106 of the inlet guide vane 100 and the outer surface 172 of the forward end portion 170 of the HP shaft 34. The seal body 186 may be positioned at least partially between the leading-edge portion 108 and the trailing-edge portion 110 of the inlet guide vane 100. The seal body 186 may be attached to the inlet guide vane 100, to the outer surface 172 of the forward end portion 170 of the HP shaft 34, or to both. In certain embodiments, the seal body 186 can be incorporated into the embodiment shown in FIG. 9, which prevents or reduces flow of the heated compressed air 184 flowing through the radial gap 112.

(85) In the embodiment of FIG. 10, the forward end portion 170 of the HP shaft 34, particularly the outer surface 172, is allowed to spin beneath the tip portion 106 of the inlet guide vane 100, breaking up and/or preventing ice buildup within the radial inlet 52 at the inlet guide vane 100 upstream from the compressor inlet 178. The seal body 186 prevents air and/or ice seepage through the radial gap 112 and into the compressor inlet 178.

(86) FIG. 11 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine 10 as shown in FIG. 1, including a portion of the inlet frame 54 and a portion of the HP compressor 22 in accordance with certain embodiments of the present disclosure. The various components shown in FIG. 11 are similar to or the same as shown in FIGS. 9-10 and described herein and are numbered likewise. In the exemplary embodiment shown in FIG. 11, the axial gap or aperture 114 is defined between the forward end portion 170 of the HP shaft 34 and the inner wall 180 of the inlet frame 54 downstream from the leading-edge portion 108 and upstream from the trailing-edge portion 110 of the inlet guide vane 100 within the radial gap 112. In other words, the forward end portion 170 of the HP shaft 34 terminates axially aft from the leading-edge portion 108 of the inlet guide vane 100 within the radial gap 112.

(87) In the embodiment shown in FIG. 11, the forward end portion 170 of the HP shaft 34, particularly the outer surface 172, is allowed to spin beneath the tip portion 106 of the inlet guide vane 100 and the purge or extraction-air source 182, such as but not limited to the HP compressor 22, provides a flow of heated compressed air as indicated by arrows 184 through the aperture 114 and directly into the radial gap 112, breaking up and/or preventing ice buildup within the radial inlet 52 at the inlet guide vane 100 upstream from the compressor inlet 178.

(88) FIG. 12 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine 10, as shown in FIG. 1, including a portion of the inlet frame 54 and a portion of the HP compressor 22 in accordance with certain embodiments of the present disclosure. The various components shown in FIG. 12 are similar to or the same as those shown in FIGS. 9-1 and described herein and are numbered likewise. FIG. 13 is a schematic aft looking forward view of the forward end portion 170 of the first rotor shaft 34 according to an embodiment of the present disclosure. As shown in FIGS. 12 and 13 collectively, the HP shaft 34 includes a plurality of protrusions 188 circumferentially spaced with respect to circumferential direction "C" (FIG. 13) about the longitudinal centerline or central axis 12 of the turboprop engine 10 and extending radially outwardly from the outer surface 172 with respect to radial direction R. As shown in FIG. 12, the plurality of protrusions 188 is disposed upstream of the leading-edge portion 108 of the inlet guide vane 100 and the radial gap 112.

(89) In the embodiment shown in FIGS. 12 and 13, the forward end portion 170 of the HP shaft 34, particularly the outer surface 172, is allowed to spin beneath the tip portion 106 of the inlet guide vane 100. The protrusions 188 break away any existing ice buildup and prevent or reduce the potential for new ice formation.

(90) FIG. 14 is an enlarged schematic cross-sectional view of a portion of the gas turbine engine 10, as shown in FIG. 1, including a portion of the inlet frame 54 and a portion of the HP

compressor **22** in accordance with certain embodiments of the present disclosure. The various components shown in FIG. **14** are similar to or the same as those shown in FIG. **9** and described herein and are numbered likewise. In the embodiment shown in FIG. **14**, the inlet guide vane **100** or more particularly, the vane body **104**, includes or defines an internal air passage **116**. The internal air passage **116** is in fluid communication with a compressed air source **190** via an inlet **192** to the internal air passage **116** for receiving a flow of compressed air as indicated by arrows **194** from the compressed air source **190**. The inlet guide vane **100** further includes or defines one or more outlets **118** defined along the tip portion **106** between the leading-edge portion **108** and the trailing-edge portion **110** of the inlet guide vane **100** within the radial gap **112**.

(91) In the embodiment shown in FIG. **14**, the forward end portion **170** of the HP shaft **34**, particularly the outer surface **172**, is allowed to spin beneath the tip portion **106** of the inlet guide vane **100**. The compressed air source **190** provides a flow of the compressed air as indicated by arrows **194** through the internal air passage **116** and out of the one or more outlets **118** directly into the radial gap **112**, melting and/or preventing ice buildup within the radial inlet **52** at the inlet guide vane **100** upstream from the compressor inlet **178**.

(92) FIG. **14** is an enlarged schematic cross-sectional view of a portion of the gas turbine engine **10**, as shown in FIG. **1**, including a portion of the inlet frame **54** and a portion of the HP compressor **22** in accordance with certain embodiments of the present disclosure. The various components shown in FIG. **15** are similar to or the same as those shown in FIGS. **9** and **14** and described herein and are numbered likewise. In the embodiment shown in FIG. **15** the inlet guide vane **100** or more particularly, the vane body **104**, includes or defines the internal air passage **116**. The internal air passage **116** is in fluid communication with the compressed air source **190** via inlet **192** to the internal air passage **116** for receiving the flow of compressed air as indicated by arrows **194** from the compressed air source **190**. The inlet guide vane **100** further includes or defines one or more outlets **120** defined along the leading-edge portion **108** proximate to the tip portion **106** of the inlet guide vane **100** upstream from the radial gap **112**.

(93) In the embodiment shown in FIG. **15**, the forward end portion **170** of the HP shaft **34**, particularly the outer surface **172**, is allowed to spin beneath the tip portion **106** of the inlet guide vane **100**. The compressed air source **190** provides a flow of the compressed air as indicated by arrows **194** through the internal air passage **116** and out of the one or more outlets **120** upstream from the radial gap **112**, melting and/or preventing ice buildup within the radial inlet **48** at the inlet guide vane **100** upstream from the compressor inlet **178**.

(94) As such, the various embodiments disclosed herein and shown in FIGS. **9-15** reduce or prevent ice accretion on the turboprop engine **10** particularly along the inner wall **180** of the inlet frame **54** at the inlet guide vane **100** due, at least in part, to the high rotational speed of the HP shaft **34**. In addition, when heated compressed air is utilized, ice or water droplets are dispersed by means of the heated compressed air coming from the purge-air source into the radial gap and/or being introduced upstream from the leading-edge portion **108** of the inlet guide vane **100**. In addition, the disclosure provides increased inlet guide vane **100** tip portion **106** temperatures by guiding the heated compressed air towards the radial gap **112**. The ability to adjust the flowrate of the heated compressed air **184** or the compressed air **194** based on ground and inflight temperature conditions can be beneficial to overall engine performance and life expectancy.

(95) Further aspects are provided by the subject matter of the following clauses:

(96) An aircraft engine assembly comprising: a gas turbine engine having a high pressure compressor, a high pressure turbine, a high pressure shaft coupling the high pressure compressor with the high pressure turbine, a low pressure turbine, and a low pressure shaft coupled to the low pressure turbine, the high pressure turbine located forward of the high pressure compressor, and the low pressure turbine located on a forward end of the gas turbine engine; an intake channel of the gas turbine engine configured to receive an incoming flow of air and thereby form an intake flow of air, the intake channel configured to turn the received incoming flow of air from an incoming

flow direction to a first axial direction of the gas turbine engine, the incoming flow direction reverse of the first axial direction; and an electric machine coupled with the low pressure shaft and located at the aft end of the gas turbine engine proximate the intake channel, the electric machine in heat exchange communication with the intake flow of air such that the electric machine transfers heat to the incoming flow of air within the intake channel when the electric machine is operated.

(97) The aircraft engine assembly of one or more of these clauses, which further includes a tail cone, and wherein the electric machine is located within the tail cone.

(98) The aircraft engine assembly of one or more of these clauses, wherein the low pressure shaft is oriented to extend through the intake channel such that an axial portion of the low pressure shaft is surrounded by the incoming flow of air in the intake channel.

(99) The aircraft engine assembly of one or more of these clauses, wherein the electric machine is positioned to receive an impingement of the intake flow of air.

(100) The aircraft engine assembly of one or more of these clauses, wherein an offtake flow of air is extracted from the intake flow of air, the offtake flow of air forming a cooling flow of air routed to the electric machine.

(101) The aircraft engine assembly of one or more of these clauses, wherein the low pressure turbine includes variable stator vanes, wherein the gas turbine engine includes an engine controller, and wherein the variable stator vanes of the low pressure turbine are controlled by the controller to change position when the electric machine changes from a first power level to a second power level.

(102) The aircraft engine assembly of one or more of these clauses, which further includes a propeller located on a forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine, and which further includes a gearbox coupled between the low pressure shaft and the propeller.

(103) The aircraft engine assembly of one or more of these clauses, which further includes a propeller located on a forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine, and wherein the propeller is variable pitch propeller, and wherein the electric machine is coaxial with the low pressure shaft.

(104) The aircraft engine assembly of one or more of these clauses, which further includes an engine nacelle enclosing the gas turbine engine, the electric machine located in an enclosed space aft of the high pressure compressor.

(105) The aircraft engine assembly of one or more of these clauses, wherein the engine nacelle includes a discharge opening sized to permit discharge of the portion of the intake flow of air.

(106) The aircraft engine assembly of one or more of these clauses, wherein the electric machine is in a conductive heat exchange communication with the intake flow of air.

(107) The aircraft engine assembly of one or more of these clauses, which further includes an engine nacelle enclosing the gas turbine engine, the electric machine located in an enclosed space aft of the high pressure compressor.

(108) The aircraft engine assembly of one or more of these clauses, wherein the engine nacelle includes a discharge opening sized to permit discharge of the portion of the intake flow of air.

(109) The aircraft engine assembly of one or more of these clauses, wherein the electric machine is in a conductive heat exchange communication with the intake flow of air.

(110) The aircraft engine assembly of one or more of these clauses, wherein the intake channel includes a first portion that directs air into a first radial side of the low pressure shaft, wherein the intake channel includes a second portion configured as annular in shape, and wherein the first portion is upstream of the second portion, the second portion directing the intake flow of air in the first axial flow direction toward the high pressure compressor.

(111) An aircraft powerplant comprising: a gas turbine engine having a high pressure compressor and a high pressure turbine, the gas turbine engine further having a high pressure shaft coupling the high pressure compressor with the high pressure turbine, the gas turbine engine also having a first

axial flow direction from the high pressure compressor to the high pressure turbine; a propeller coupled to a low pressure turbine of the gas turbine engine using a low pressure shaft, the low pressure shaft located coaxial with the high pressure shaft, the propeller configured to receive a free stream flow of air oriented in a freestream direction and impart work upon the free stream flow of air, the propeller located on an upstream side of the freestream direction from the high pressure turbine; an intake channel defining an intake flow of air in fluid communication with the gas turbine engine, the intake channel configured to reverse the intake flow of air initially flowing in the freestream direction to the first axial flow direction of the gas turbine engine; and an electric machine coupled to the low pressure shaft and located on an opposite side of the gas turbine engine from the propeller, the electric machine positioned to be cooled by a portion of the intake flow of air defined by the intake channel.

(112) The aircraft powerplant of one or more of these clauses, which further includes an engine nacelle enclosing the gas turbine engine, the electric machine located in an enclosed space aft of the high pressure compressor.

(113) The aircraft powerplant of one or more of these clauses, wherein the engine nacelle includes a discharge opening sized to permit discharge of the portion of the intake flow of air.

(114) The aircraft powerplant of one or more of these clauses, wherein the electric machine is in a conductive heat exchange communication with the intake flow of air.

(115) The aircraft powerplant of one or more of these clauses, wherein the intake channel includes a first portion that directs air into a first radial side of the low pressure shaft, wherein the intake channel includes a second portion configured as annular in shape, and wherein the first portion is upstream of the second portion, the second portion directing air in the first axial flow direction toward the high pressure compressor.

(116) The aircraft powerplant of one or more of these clauses, wherein the gas turbine engine includes an engine controller, and wherein the engine controller commands a change in fuel flow rate to a combustion section of the gas turbine engine when the electric machine changes from a first power level to a second power level.

(117) The aircraft powerplant of one or more of these clauses, wherein the low pressure turbine is a free turbine.

(118) The aircraft powerplant of one or more of these clauses, an engine exhaust configured to receive an exhaust flow in the first axial flow direction from the low pressure turbine and discharge the exhaust flow having a second axial direction component reverse of the first axial direction.

(119) A method of cooling an electric machine comprising: operating a gas turbine engine having a high pressure compressor, a high pressure turbine, and a high pressure shaft, the high pressure compressor coupled to the high pressure turbine via the high pressure shaft, the gas turbine engine also including a low pressure turbine coupled via a low pressure shaft with a propeller; receiving an incoming flow of air in an incoming flow of air direction into an intake channel of the gas turbine engine, the incoming flow of air used in a combustion process of the gas turbine engine, the intake channel forming an intake channel flow path and defining an intake flow air; turning the intake flow of air using the intake channel from the incoming flow of air direction to an axial flow direction of the gas turbine engine; and cooling an electric machine located aft of the low pressure turbine using the intake flow of air flowing through the intake channel flow path.

(120) The method of cooling an electric machine of one or more of these clauses, wherein the cooling includes impingement cooling the electric machine using the intake flow of air.

(121) The method of cooling an electric machine of one or more of these clauses, wherein turning the flow includes changing shape of the flow from a non-annular shape to an annular shape.

(122) The method of cooling an electric machine of one or more of these clauses, wherein the electric machine is located in a tail cone located aft of the gas turbine engine.

(123) An aircraft engine assembly comprising: a gas turbine engine having a high pressure compressor, a high pressure turbine, a high pressure shaft coupling the high pressure compressor

with the high pressure turbine, a low pressure turbine, and a low pressure shaft coupled to the low pressure turbine, the high pressure turbine located forward of the high pressure compressor, and the low pressure turbine located on a forward end of the gas turbine engine; a propeller located on a forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine; an intake channel of the gas turbine engine configured to receive an incoming flow of air and form an intake flow of air, the intake channel configured to turn the received incoming flow of air from an incoming flow direction to a first axial direction of the gas turbine engine, the incoming flow direction reverse of the first axial direction; and an electric machine coupled with the low pressure shaft and located on a side of the high pressure compressor opposite of the high pressure turbine and proximate the intake channel, the electric machine in heat exchange communication with the intake flow of air such that the electric machine transfers heat to the incoming flow of air within the intake channel when the electric machine is operated. The high pressure compressor includes: a shaft comprising a forward end portion, wherein the forward end portion defines an outer surface and rotates with the shaft; a first row of compressor rotor blades coupled to the shaft downstream from the forward end portion; an outer casing at least partially surrounding the first row of compressor rotor blades and the outer surface of the forward end portion of the shaft, the outer casing at least partially defining an inlet to the high pressure compressor; and an inlet guide vane comprising a mounting portion, a tip portion, a leading-edge portion, and a trailing-edge portion, wherein the mounting portion is coupled to the outer casing upstream from the first row of compressor rotor blades, wherein the tip portion extends towards the outer surface of the forward end portion of the shaft, and wherein a radial gap is defined between the tip portion and the outer surface.

(124) The aircraft engine assembly of any preceding clause, wherein the forward end portion of the shaft and an inner wall of an inlet frame upstream from the inlet guide vane define an aperture therebetween, wherein the aperture is in fluid communication with a purge-air source, wherein the aperture is oriented to direct a flow of purge-air towards the radial gap.

(125) The aircraft engine assembly of any preceding clause, wherein the forward end portion of the shaft terminates axially forward of the leading-edge portion of the inlet guide vane.

(126) The aircraft engine assembly of any preceding clause, further comprising a seal body disposed within the radial gap.

(127) The aircraft engine assembly of any preceding clause, wherein the forward end portion of the shaft and an inner wall of an inlet frame define an aperture therebetween, wherein the aperture is located between the leading-edge portion and the trailing-edge portion of the inlet guide vane, wherein the aperture is in fluid communication with a purge-air source, and wherein the aperture is oriented to direct a flow of purge air into the radial gap.

(128) The aircraft engine assembly of any preceding clause, wherein the forward end portion of the rotor shaft terminates axially aft from the leading-edge portion of the inlet guide vane.

(129) The aircraft engine assembly of any preceding clause, wherein the forward end portion of the shaft includes a plurality of protrusions circumferentially spaced about and extending radially outwardly from the outer surface, wherein the plurality of protrusions is disposed upstream of the radial gap.

(130) The aircraft engine assembly of any preceding clause, wherein the inlet guide vane includes an internal air passage having an inlet in fluid communication with a purge air source and an outlet defined along the leading-edge portion of the inlet guide vane upstream from the radial gap.

(131) The aircraft engine assembly of any preceding clause, wherein the inlet guide vane includes an internal air passage having an inlet in fluid communication with a purge air source and an outlet defined within the radial gap.

(132) The aircraft engine assembly of any preceding clause, wherein the outer casing includes an inlet frame, wherein the mounting portion of the inlet guide vane is coupled to the inlet frame.

(133) A turboprop aircraft powerplant comprising: a gas turbine engine having a high pressure

compressor and a high pressure turbine, the gas turbine engine further having a high pressure shaft coupling the high pressure compressor with the high pressure turbine, the gas turbine engine also having a first axial flow direction from the high pressure compressor to the high pressure turbine; a propeller coupled to a low pressure turbine of the gas turbine engine using a low pressure shaft, the low pressure shaft located coaxial with the high pressure shaft, the propeller configured to receive a free stream flow of air oriented in a freestream direction and impart work upon the free stream flow of air, the propeller located on an upstream side of the freestream direction from the high pressure turbine; an intake channel defining an intake flow of air in fluid communication with the gas turbine engine, the intake channel configured to reverse the intake flow of air initially flowing in the freestream direction to the first axial flow direction of the gas turbine engine; and an electric machine coupled to the low pressure shaft and located on an opposite side of the gas turbine engine from the propeller, the electric machine positioned to be cooled by a portion of the intake flow of air defined by the intake channel. The high pressure compressor includes: a shaft comprising a forward end portion, wherein the forward end portion defines an outer surface and rotates with the shaft; a first row of compressor rotor blades coupled to the shaft downstream from the forward end portion; an outer casing at least partially surrounding the first row of compressor rotor blades and the outer surface of the forward end portion of the shaft, the outer casing at least partially defining an inlet to the high pressure compressor; and an inlet guide vane comprising a mounting portion, a tip portion, a leading-edge portion, and a trailing-edge portion, wherein the mounting portion is coupled to the outer casing upstream from the first row of compressor rotor blades, wherein the tip portion extends towards the outer surface of the forward end portion of the shaft, and wherein a radial gap is defined between the tip portion and the outer surface.

(134) The turboprop aircraft powerplant of any preceding clause, wherein the forward end portion of the shaft and an inner wall of an inlet frame upstream from the inlet guide vane define an aperture therebetween, wherein the aperture is in fluid communication with a purge-air source, wherein the aperture is oriented to direct a flow of purge-air towards the radial gap.

(135) The turboprop aircraft powerplant of any preceding clause, wherein the forward end portion of the shaft terminates axially forward of the leading-edge portion of the inlet guide vane.

(136) The turboprop aircraft powerplant of any preceding clause, further comprising a seal body disposed within the radial gap.

(137) The turboprop aircraft powerplant of any preceding clause, wherein the forward end portion of the shaft and an inner wall of an inlet frame define an aperture therebetween, wherein the aperture is located between the leading-edge portion and the trailing-edge portion of the inlet guide vane, wherein the aperture is in fluid communication with a purge-air source, and wherein the aperture is oriented to direct a flow of purge air into the radial gap.

(138) The turboprop aircraft powerplant of any preceding clause, wherein the forward end portion of the rotor shaft terminates axially aft from the leading-edge portion of the inlet guide vane.

(139) The turboprop aircraft powerplant of any preceding clause, wherein the forward end portion of the shaft includes a plurality of protrusions circumferentially spaced about and extending radially outwardly from the outer surface, wherein the plurality of protrusions is disposed upstream of the radial gap.

(140) The turboprop aircraft powerplant of any preceding clause, wherein the inlet guide vane includes an internal air passage having an inlet in fluid communication with a purge air source and an outlet defined along the leading-edge portion of the inlet guide vane upstream from the radial gap.

(141) The turboprop aircraft powerplant of any preceding clause, wherein the inlet guide vane includes an internal air passage having an inlet in fluid communication with a purge air source and an outlet defined within the radial gap.

(142) The turboprop aircraft powerplant of any preceding clause, wherein the outer casing includes an inlet frame, wherein the mounting portion of the inlet guide vane is coupled to the inlet frame.

(143) This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. An aircraft engine assembly comprising: a gas turbine engine having a high pressure compressor, a high pressure turbine, a high pressure shaft coupling the high pressure compressor with the high pressure turbine, a low pressure turbine, and a low pressure shaft coupled to the low pressure turbine, the high pressure turbine located forward of the high pressure compressor, and the low pressure turbine located on a forward end of the gas turbine engine; a propeller located on a forward end of the gas turbine engine and coupled via the low pressure shaft with the low pressure turbine; an intake channel of the gas turbine engine configured to receive an incoming flow of air and form an intake flow of air, the intake channel configured to turn the received incoming flow of air from an incoming flow direction to a first axial direction of the gas turbine engine, the incoming flow direction reverse of the first axial direction; and an electric machine coupled with the low pressure shaft and located on a side of the high pressure compressor opposite of the high pressure turbine and proximate the intake channel, the electric machine in heat exchange communication with the intake flow of air such that the electric machine transfers heat to the incoming flow of air within the intake channel when the electric machine is operated, wherein the high pressure compressor includes: a shaft comprising a forward end portion, wherein the forward end portion defines an outer surface and rotates with the shaft; a first row of compressor rotor blades coupled to the shaft downstream from the forward end portion; an outer casing at least partially surrounding the first row of compressor rotor blades and the outer surface of the forward end portion of the shaft, the outer casing at least partially defining an inlet to the high pressure compressor; and an inlet guide vane comprising a mounting portion, a tip portion, a leading-edge portion, and a trailing-edge portion, wherein the mounting portion is coupled to the outer casing upstream from the first row of compressor rotor blades, wherein the tip portion extends towards the outer surface of the forward end portion of the shaft, and wherein a radial gap is defined between the tip portion and the outer surface.
2. The aircraft engine assembly of claim 1, wherein the forward end portion of the shaft and an inner wall of an inlet frame upstream from the inlet guide vane define an aperture therebetween, wherein the aperture is in fluid communication with a purge-air source, wherein the aperture is oriented to direct a flow of purge-air towards the radial gap.
3. The aircraft engine assembly of claim 1, wherein the forward end portion of the shaft terminates axially forward of the leading-edge portion of the inlet guide vane.
4. The aircraft engine assembly of claim 1, further comprising a seal body disposed within the radial gap.
5. The aircraft engine assembly of claim 1, wherein the forward end portion of the shaft and an inner wall of an inlet frame define an aperture therebetween, wherein the aperture is located between the leading-edge portion and the trailing-edge portion of the inlet guide vane, wherein the aperture is in fluid communication with a purge-air source, and wherein the aperture is oriented to direct a flow of purge air into the radial gap.
6. The aircraft engine assembly of claim 1, wherein the forward end portion of the rotor shaft terminates axially aft from the leading-edge portion of the inlet guide vane.

7. The aircraft engine assembly of claim 1, wherein the forward end portion of the shaft includes a plurality of protrusions circumferentially spaced about and extending radially outwardly from the outer surface, wherein the plurality of protrusions is disposed upstream of the radial gap.
8. The aircraft engine assembly of claim 1, wherein the inlet guide vane includes an internal air passage having an inlet in fluid communication with a purge air source and an outlet defined along the leading-edge portion of the inlet guide vane upstream from the radial gap.
9. The aircraft engine assembly of claim 1, wherein the inlet guide vane includes an internal air passage having an inlet in fluid communication with a purge air source and an outlet defined within the radial gap.
10. The aircraft engine assembly of claim 1, wherein the outer casing includes an inlet frame, wherein the mounting portion of the inlet guide vane is coupled to the inlet frame.
11. A turboprop aircraft powerplant comprising: a gas turbine engine having a high pressure compressor and a high pressure turbine, the gas turbine engine further having a high pressure shaft coupling the high pressure compressor with the high pressure turbine, the gas turbine engine also having a first axial flow direction from the high pressure compressor to the high pressure turbine; a propeller coupled to a low pressure turbine of the gas turbine engine using a low pressure shaft, the low pressure shaft located coaxial with the high pressure shaft, the propeller configured to receive a free stream flow of air oriented in a freestream direction and impart work upon the free stream flow of air, the propeller located on an upstream side of the freestream direction from the high pressure turbine; an intake channel defining an intake flow of air in fluid communication with the gas turbine engine, the intake channel configured to reverse the intake flow of air initially flowing in the freestream direction to the first axial flow direction of the gas turbine engine; and an electric machine coupled to the low pressure shaft and located on an opposite side of the gas turbine engine from the propeller, the electric machine positioned to be cooled by a portion of the intake flow of air defined by the intake channel, wherein the high pressure compressor includes: a shaft comprising a forward end portion, wherein the forward end portion defines an outer surface and rotates with the shaft; a first row of compressor rotor blades coupled to the shaft downstream from the forward end portion; an outer casing at least partially surrounding the first row of compressor rotor blades and the outer surface of the forward end portion of the shaft, the outer casing at least partially defining an inlet to the high pressure compressor; and an inlet guide vane comprising a mounting portion, a tip portion, a leading-edge portion, and a trailing-edge portion, wherein the mounting portion is coupled to the outer casing upstream from the first row of compressor rotor blades, wherein the tip portion extends towards the outer surface of the forward end portion of the shaft, and wherein a radial gap is defined between the tip portion and the outer surface.
12. The turboprop aircraft powerplant of claim 11, wherein the forward end portion of the shaft and an inner wall of an inlet frame upstream from the inlet guide vane define an aperture therebetween, wherein the aperture is in fluid communication with a purge-air source, wherein the aperture is oriented to direct a flow of purge-air towards the radial gap.
13. The turboprop aircraft powerplant of claim 11, wherein the forward end portion of the shaft terminates axially forward of the leading-edge portion of the inlet guide vane.
14. The turboprop aircraft powerplant of claim 11, further comprising a seal body disposed within the radial gap.
15. The turboprop aircraft powerplant of claim 11, wherein the forward end portion of the shaft and an inner wall of an inlet frame define an aperture therebetween, wherein the aperture is located between the leading-edge portion and the trailing-edge portion of the inlet guide vane, wherein the aperture is in fluid communication with a purge-air source, and wherein the aperture is oriented to direct a flow of purge air into the radial gap.
16. The turboprop aircraft powerplant of claim 11, wherein the forward end portion of the rotor shaft terminates axially aft from the leading-edge portion of the inlet guide vane.
17. The turboprop aircraft powerplant of claim 11, wherein the forward end portion of the shaft

includes a plurality of protrusions circumferentially spaced about and extending radially outwardly from the outer surface, wherein the plurality of protrusions is disposed upstream of the radial gap.

18. The turboprop aircraft powerplant of claim 11, wherein the inlet guide vane includes an internal air passage having an inlet in fluid communication with a purge air source and an outlet defined along the leading-edge portion of the inlet guide vane upstream from the radial gap.

19. The turboprop aircraft powerplant of claim 11, wherein the inlet guide vane includes an internal air passage having an inlet in fluid communication with a purge air source and an outlet defined within the radial gap.

20. The turboprop aircraft powerplant of claim 11, wherein the outer casing includes an inlet frame, wherein the mounting portion of the inlet guide vane is coupled to the inlet frame.
